MILITARY SPECIFICATION

HOOD AND SOCKS, EXTREME COLD WEATHER SLEEPING SYSTEM (ECWSS)

This specification is approved for use by the U.S. Army Natick, Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a polyester fiberpile hood and a pair of polyester fiberpile socks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage: General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets
MIL-STD-1491 - Glossary of Knitting Imperfections

DRAWINGS

U. S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-2-451 - Hood, Extreme Cold Weather Sleeping System
2-2-454 - Socks, Extreme Cold Weather Sleeping System

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample in which case this document shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials. The materials shall be as specified herein and on Drawings 2-2-451 and 2-2-454 and all subsidiary drawings and parts lists pertaining thereto.

3.3.1 Labels. Each hood and pair of socks shall have a class 15 combination identification and instruction label conforming to type VI of DDD-L-20. The label color shall be Medium Green, C.A. 70034, 70130, or 70131 and shall show fastness to laundering. The label shall read as follows, stating only the appropriate item (hood or socks):

HOOD, SOCKS, EXTREME COLD WEATHER SLEEPING SYSTEM

a. Field Laundry: The hood/socks should be laundered utilizing "Formula II" of TM 10-280. NOTE: DO NOT USE CHLORINE BLEACH, STARCH, DRY CLEAN.

Drying: The hood/socks should be finished in a tumble-dryer, loaded not more than 2/3 of rated capacity at lowest fabric cycle (do not exceed 90°F). Remove immediately from dryer. Avoid over drying. NOTE: DO NOT PRESS.

b. Machine/Hand Laundering. Use delicate/gentle fabric cycle or by hand, using cold water (up to 85°F) and cold water laundry detergent. Rinse in clean cold water. NOTE: DO NOT USE CHLORINE BLEACH OR STARCH.

Drying: Tumble dry at lowest fabric cycle, delicate/gentle (do not exceed 90°F). Remove immediately at end of drying. Avoid over-drying. To drip dry, remove from water and place on rust-proof hanger. NOTE: DO NOT PRESS.

DO NOT REMOVE THIS LABEL

3.4 Design and construction. The design and construction of the hood and socks shall be as specified herein and as shown on Drawings 2-2-451 and 2-2-454 and all subsidiary drawings and parts lists pertaining thereto.

3.4.1 Stitches, seams and stitchings. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as shown on Drawings 2-2-451 and 2-2-454 and all subsidiary drawings and parts lists pertaining thereto. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur. Overedge stitching shall be 3/16 to 1/4 inch gauge.

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3.4.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Ends of continuous line of stitching shall be overlapped not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excess-ively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.1.2 Types 503 and 504 stitching. All repairs shall be in accordance with 3.4.1.1.1.a. and 3.4.1.1.1.b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn.

3.4.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.5 Finished measurements.

3.5.1 Finished measurements (HOOD). The hood shall conform to the following finished measurements:

Area to be measured	Measurement inches	Tolerance inch
Length 1/	13	± 1
Widest 2/	11-1/4	± 3/4
Face opening 3/	11	± 1/2

1/ With the drawstring totally loosened place hood face side up on a flat surface. Align bottom edges of front and back of bib. Fold top half of

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face opening flat approximating a straight horizontal line. With hood in this position measure length in a straight line from the folded edge of the crown at the highest point to the bottom of the hood at the lowest point on the bib.

2/ With the hood positioned as indicated at footnote 1/ measure across the hood in a straight line along the top of the face opening from folded edge to folded edge.

3/ Fold hood in half lengthwise so the fold runs in the length direction of the center top cover and the front center neck seam. Place the folded hood on a flat surface with the edges of the face opening even. Place one hand on the neck end of the face opening, holding the hood against the flat surface. With the other hand slide the top of the hood along the flat surface toward the back of the hood until the curvature along the side edges of the face opening approximate a straight line. With the hood in this position measure along the edge of the face opening from the fold at the center of the front neck seam to the fold at the center of the top cover.

3.5.1 Finished measurements (SOCK). The sock shall conform to the following finished measurements:

Area to be measured	Measurement inches	Tolerance inch
Height 1/	11-1/2	± 1
Cuff opening 2/	5	± 1/2
Sole 3/	12	± 3/4

1/ Place the sock on a flat surface and fold it in half lengthwise so that the folds run lengthwise along the nylon bottom, the back center heel and the front center toe cap. With the sock in this position measure height in a straight line from the top cuff to the fold in the nylon sole.

2/ With the sock positioned as indicated at footnote 1/ measure horizontally across the top cuff from folded edge to folded edge.

3/ With the sock positioned as indicated at footnotes 1/ and 2/ measure along the nylon sole seam horizontally from the folded edge of the toe to the folded edge of the heel.

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3.6 Workmanship. The finished hood and socks shall conform to the quality of products established by this document. The occurrence of defects shall not exceed the applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the requirements for barrel locks on Drawing 2-2-451 and subsidiary drawings pertaining thereto.

4.4.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of hoods or socks. The sample unit shall be one hood or one pair of socks. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Material defects	Hole, cut, tear, smash, run or dropped stitch 1/	X	
	Needle chew	X	
	Thin or uneven area	X	
	Mend 1/	X	
	Slub (more than twice the size of the normal diameter of the yarn), float or snag 1/		X
Color and shade	Color not as specified	X	
	Variation in shade within a part or between two or more parts 2/ 3/		X
	Streaks clearly visible 3/		X
	Barre 1/		X
Cleanness	Spot or stain more than 1/4 inch in largest dimension		X
	Two or more thread ends not trimmed to 1/2 inch or less in length		X
	One or more shade tickets not removed		X
	One or more size or shade markings clearly visible on outside 3/		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein)	X	
	Any operation not performed as specified (unless otherwise classified herein)		X
Cutting	Any component part not cut in accordance with directional lines as indicated on patterns or in accordance with document requirements	X	
Seams and stitching	Any open seam (except on hems)	X	
	Open seam on hems:		
	- 1/4 to 1 inch		X
	- over 1 inch	X	
NOTE:	A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Seam or stitch type not as specified	X	
	Parts of hood/socks badly pleated, caught or twisted in any type of stitching	X	
	Seam twisted or puckered		X
	Thread breaks (all stitch types) backstitched less than 1/2 inch		X
	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/2 inch		X
	Ends of a continuous line of stitching overlapping less than 1/2 inch		X
	Raw edges:		
	Any on outside	X	
	On inside 1/8 to 1/2 inch (inclusive)		X
	On inside, over 1/2 inch	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Runoff, or any line of stitching not catching all plies of material to be seamed:		
	- up to 1/2 inch (inclusive)		X
	- more than 1/2 inch	X	
	Stitch tension:		
	Loose tension resulting in a loose seam	X	
	Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X	
	Two or more stitches per inch less than specified minimum or more than specified maximum	X	
Hems	Omitted	X	
Label	Missing, incorrect, illegible, misplaced, or not attached as specified	X	

1/ For knitted fabric imperfection defects, see MIL-STD-1491.

2/ All parts shall be examined with wales and courses running in same direction.

3/ At normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The end items shall be examined for the finished dimensions specified in 3.5.1 and 3.5.2. Any dimension not within the specified tolerances shall be scored as a defect. The lot size shall be expressed in units of hoods or socks. The sample unit shall be one hood or one pair of socks. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. One hood and one pair of socks shall be bundled together and securely tied at each end with cotton tape or twine to form a unit pack. The dimensions of the bundle shall be approximately 12 inches in length, 4 inches in width, and 4 inches in depth.

5.1.2 Commercial. One hood and one pair of socks shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twenty-four bundles, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of the box shall be approximately 24 inches in length, 12 inches in width, and 16 inches in depth. Each shipping container shall be closed in accordance with method III, waterproofed in accoradance with method V, and reinforced in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twenty-four bundles, preserved as specified in 5.1, shall be packed in a shipping container. The shipping container shall be a fiberboard box conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Inside dimensions of the box shall be approximately 24 inches in length, 12 inches in width, and 16 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Bundles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), bundles, packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 35 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The hood and socks are intended for use by personnel of the Department of Defense as a component of the Extreme Cold Weather Sleeping System.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article sample is required (see 3.2, 4.3, and 6.4).
c. Selection of applicable levels of preservation, and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Subject term (key word) listing.

ECWSS
Hood
Socks

Custodians:

Army - GL

Preparing activity:

Army - GL

Project No. 8465-0990

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-44308(GL)	Hood and Socks, Extreme Cold Weather Sleeping System (ECWSS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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